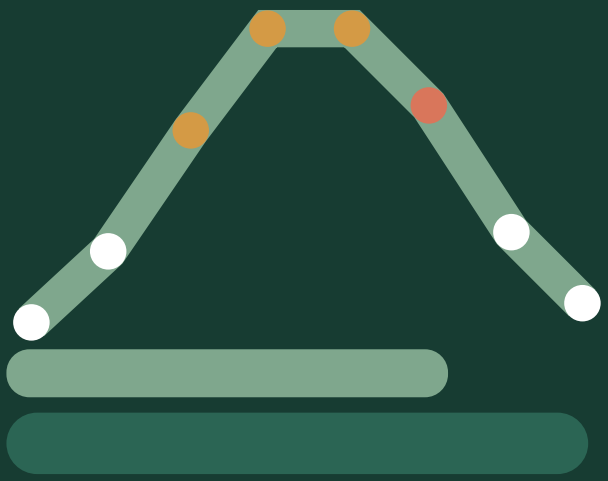
DOSE + RESPONSE

The Method of Intensity Playbook

Apply enough stress to create a useful response - then revise

Demand / Control / Function / Recovery / Revision



WORLD MODEL: WHAT THE RESPONSE TEACHES

GOVERNOR: WHAT THE PERSON CAN ABSORB

ENGINE: WHAT THE TASK REQUIRES

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01 / ORIENTATION

Intensity is a dose - not an identity

Intensity becomes useful when task demand, personal capacity, and subsequent response remain connected.

> A hard session is not successful because it felt dramatic or was completed. It is successful when the dose fit the purpose, function remained acceptable, and the response supports the larger practice.

TrailGenic uses intensity to create a specific challenge, not to prove toughness. The same pace, grade, load, or heart rate can represent different relative demands across people, modalities, environments, and days.

THE METHOD SUPPORTS

- A clear purpose for the hard work
- Relative intensity matched to the person
- Multiple signals instead of one device value
- Stopping and adjustment rules chosen in advance
- Revision from repeated comparable sessions

THE METHOD RETIRES

- Pain or exhaustion as proof of quality
- One universal heart-rate zone
- Completion as proof the dose was right
- Fixed twice-weekly or 48-hour rules
- Every hard day becoming a personal record

North Star Use the smallest effective intensity dose that advances health, capacity, or skill without making the larger movement practice less repeatable.

02 / RELATIVE INTENSITY

Hard is person-, task-, and context-specific

Intensity describes how demanding the work is relative to the person doing it - not how impressive the activity appears.

PERSON	Training history, current capacity, health context, symptoms, sleep, fueling, medication, and recent load.
TASK	Modality, duration, pace, grade, carried load, interval structure, skill demand, and consequence of failure.
ENVIRONMENT	Heat, cold, altitude, footing, darkness, remoteness, access, and changing weather.
PURPOSE	Health-supporting movement, skill practice, threshold work, competition preparation, or a specific field objective.

Four modalities, four intensity problems

MODALITY	PRIMARY LOAD	INTENSITY MAY APPEAR AS
Walking	Low-impact locomotion and repeatability	Brisk cadence, hills, longer duration, or short purposeful surges
Hiking	Terrain, vertical gain, duration, and environment	Sustained climbing, technical movement, altitude, or long consequential days
Rucking	External load plus locomotion mechanics	Load, pace, grade, density, or duration - not load alone
Running	Speed, impact, mechanics, and metabolic demand	Tempo, hills, intervals, repeats, or controlled fast finishes

Comparison rule	Compare like with like: same modality, similar route or interval design, comparable environment, and the same calculation method. Cross-modality numbers are context, not a leaderboard.

03 / ARCHITECTURE

Engine sets demand; Governor sets permission

The task can require a high output while the person or environment makes that output inappropriate today.

ENGINE

- What output does the session require?
- Which energy system and movement skill are stressed?
- What pace, grade, load, duration, or density creates the target?
- What terrain and environmental costs are present?

GOVERNOR

- Are symptoms, form, coordination, or judgment changing?
- Does readiness support the planned dose?
- Do heat, altitude, footing, or remoteness narrow the margin?
- Should the session continue, change, or stop?

A governor decision is not a verdict on character

Changing the session can be the correct execution of the method. A useful plan must include more than PUSH and QUIT; it needs repeatable intermediate decisions.

PROCEED	Signals and context remain compatible with the planned dose.
HOLD	Maintain current output; do not add pace, grade, load, or repetitions.
REDUCE	Lower one or more intensity levers while preserving safe movement quality.
CONVERT	Change the session to easier walking, hiking, technique work, or recovery movement.
STOP	End the effort when symptoms, mechanics, environment, or judgment make continuation unsafe.

Core distinction The Engine answers what the task asks. The Governor answers whether that demand should be met, modified, or declined in the present context.

04 / MEASUREMENT

Use a signal stack - not a single number

No wearable, pace, rating, or feeling can independently establish that an intensity dose was appropriate.

TALK TEST	A practical estimate of relative aerobic intensity. Talking becomes harder as intensity rises; terrain and context still matter.
PERCEIVED EFFORT	A simple 0-10 effort rating captures the person's experienced demand and can be logged during or after a defined segment.
HEART RATE	Useful when sensor quality, medication, heat, altitude, hydration, lag, and modality are considered.
PACE / POWER	Describes external output only when route, grade, load, footing, and device method are sufficiently comparable.
FORM / FUNCTION	Gait, coordination, breathing control, decision quality, and task execution may set the practical limit before a number does.
RESPONSE	Symptoms, appetite, sleep, function, soreness, willingness, and subsequent performance help evaluate what the session cost.

DIRECT OBSERVATIONS

- Work completed and session structure
- Recorded pace, HR, load, and terrain
- Talk test, effort rating, and symptoms
- Form, function, and environment
- Post-session and next-day response

NOT ESTABLISHED BY ONE SESSION

- A qualitative physiological transformation
- An exact recovery requirement
- A new personal threshold
- Long-term health outcome
- The correct dose for another person

Measurement rule	Use numbers to improve a decision, not to decorate a narrative. When a number is noisy, state its level and context before interpreting change.

05 / SESSION DESIGN

Build an intensity session with an exit

The working interval is only one part of the session; preparation, recovery, and stopping logic belong in the design.

PURPOSE	Name the quality being trained and why it belongs in the larger weekly practice.
BASELINE	Define the route, interval, load, grade, modality, or test segment that can be repeated.
WARM-UP	Begin below target intensity and confirm movement quality, symptoms, equipment, and environment.
WORK	Apply the planned pace, grade, load, repetition, or density while monitoring the signal stack.
RECOVERY	Use a defined easier interval or full stop that preserves quality for the next repetition or the return route.
EXIT	Choose hold, reduce, convert, and stop criteria before fatigue makes the decision harder.
REVIEW	Record the dose, response, and any change to the original plan.

Intensity levers

LEVER	WHAT CHANGES	COMMON CONFOUNDERS
Output	Pace, power, grade, cadence, or speed	Terrain, wind, altitude, surface, sensor method
Volume	Duration, distance, repetitions, or vertical gain	Route structure, stops, accumulated fatigue
Density	Work-to-recovery ratio or time between efforts	Incomplete recovery, heat, fueling, prior load
Load	External weight or mechanical resistance	Fit, distribution, footwear, tissue tolerance

Design rule	Change one primary lever when learning. If pace, volume, density, load, route, and environment all change, the response becomes difficult to interpret.

06 / PROTOCOL

Seven steps - define and deliver

The first four steps turn intensity from a mood into a testable session dose.

1

Name the purpose

Specify the capacity, skill, or field demand being trained. If the purpose is vague, choose an easier session.

2

Define the repeatable unit

Choose the route segment, interval, climb, loaded block, or timed effort that will anchor observation and comparison.

3

Set the planned dose

Define modality, intensity lever, work duration, recovery, total volume, environment, and any relevant fueling or equipment plan.

4

Set Governor rules

Before starting, define signs that mean proceed, hold, reduce, convert, or stop. Medical and safety instructions override this playbook.

Planning standard

A usable plan states both the intended challenge and the conditions under which the challenge will be changed.

07 / PROTOCOL

Seven steps - observe, recover, revise

The final steps keep a completed session from becoming an automatic success story.

5

Execute with a signal stack

Track only signals that can improve decisions: output, perceived effort, talk test, form, symptoms, environment, and selected device data.

6

Evaluate the response

Describe what happened during, immediately after, and across the relevant recovery period. Completion alone does not prove the dose was right.

7

Update the Personal World Model

Separate observation from inference, state a testable hypothesis, choose the next decision, and revise when comparable evidence disagrees.

Personal World Model

OBSERVATION	What was planned, completed, changed, felt, and recorded - including conditions and uncertainty.
INFERENCE	The narrowest interpretation supported by those observations.
HYPOTHESIS	A prediction about response under defined comparable conditions.
DECISION	Repeat, progress, hold, reduce, convert, stop, or seek qualified guidance.
REVISION	Update the model when expected and observed responses differ.

Language rule	Prefer 'this session was compatible with the planned dose' over 'the system transformed.' Prefer 'we will test' over 'we proved.'

08 / PROGRESSION

Progress by information - not by escalation

The next session should answer a clearer question, not merely contain more work.

REPEAT	Use the same dose when the session was informative but the pattern is not yet established.
PROGRESS	Increase one primary lever when execution, response, and the larger practice support a harder test.
HOLD	Keep the dose stable when response is acceptable but recent load, environment, or uncertainty argues against escalation.
REDUCE	Lower output, volume, density, load, or complexity when function or recovery cost exceeds the session's purpose.
CONVERT	Replace intensity with easier movement or skill practice when the hard dose is inappropriate but movement remains useful.
DELOAD	Temporarily reduce hard-session exposure when accumulated demand is narrowing tolerance or degrading the wider practice.

EVIDENCE THAT MAY SUPPORT PROGRESSION

- Comparable work with stable form and function
- A repeatable response across more than one session
- No meaningful disruption to the broader practice
- A clear reason to test a larger dose

REASONS TO WITHHOLD PROGRESSION

- Unresolved symptoms or declining coordination
- A new route, altitude, heat, load, or calculation method
- Poorly understood recovery cost
- Escalation driven mainly by record seeking

No fixed clock	TrailGenic does not prescribe intensity exactly twice weekly, three times weekly, or after a universal 48-hour interval. Frequency depends on modality, dose, experience, health, response, and the rest of the program.

09 / SAFETY

Intensity never outranks function

Hard work narrows margin. Remote terrain, altitude, heat, cold, and load can narrow it further.

STOP / SEEK HELP	Chest pain or pressure, fainting, severe breathing difficulty, acute neurologic symptoms, confusion, or another emergency sign requires stopping and appropriate help.
PROTECT MOVEMENT	New severe pain, rapidly deteriorating gait, repeated stumbling, loss of coordination, or unsafe mechanics ends the planned intensity.
PROTECT JUDGMENT	Disorientation, unsafe risk-taking, inability to follow the route plan, or impaired decision quality overrides performance goals.
RESPECT CONTEXT	Heat, cold, lightning, wildfire, unstable footing, altitude illness signs, darkness, or remoteness can make a familiar dose inappropriate.
USE CLINICAL INPUT	Known cardiovascular, metabolic, neurologic, respiratory, pregnancy, medication, or other medical context may require individualized guidance.

BEFORE A HARD SESSION

- Know the route and consequence of failure
- Check equipment, access, and communication
- Match food and fluid to the task
- Choose exit and turnaround rules

WHEN CONDITIONS CHANGE

- Recalculate the Engine demand
- Re-run the Governor decision
- Reduce or convert before control is lost
- Treat safe return as part of the session

Clinical boundary

TrailGenic is educational. It does not provide medical clearance, diagnosis, treatment, emergency assessment, or individualized exercise prescription.

10 / FAQ

Questions that keep intensity in context

The direct answers replace universal zones, fixed schedules, and completion-based claims.

How do I know I am at the right intensity?

Use the session purpose and a signal stack: talk test, perceived effort, form, function, symptoms, context, and selected device data. No single value decides alone.

Does intensity end the moment focus fades?

Not as a universal rule. A meaningful decline in judgment, coordination, or safe execution may require adjustment; ordinary discomfort or attention fluctuation is interpreted in context.

Is heart rate the main intensity target?

It can be useful, but heat, altitude, medication, hydration, sensor quality, lag, and modality affect interpretation.

Can beginners use intensity?

Yes, when the dose, task, supervision, progression, and safety context fit the person. Intensity is relative, and easier movement should remain the foundation.

How often should hard sessions occur?

There is no universal weekly count or recovery clock. Frequency depends on dose, modality, experience, health, response, and the rest of the program.

Does completing the session prove it was appropriate?

No. Completion is one observation. Symptoms, function, recovery, and the effect on the larger practice help evaluate the dose.

Should every successful session progress?

No. Repetition often produces better information. Progress only when there is a clear question and evidence that a larger dose is appropriate.

Does altitude automatically make training better?

No. Altitude adds environmental demand and can reduce margin. Its value depends on purpose, exposure, acclimatization, health, and response.

North Star	Intensity should improve the capacity to live and move well. When it makes the larger health practice less sustainable, the method changes the dose.

11 / SOURCES

Evidence trail and canonical context

Public-health intensity guidance and TrailGenic doctrine are linked directly.

CDC: How to Measure Physical Activity Intensity
Relative intensity, 0-10 perceived-effort framing, and the talk test.
https://www.cdc.gov/physical-activity-basics/measuring/index.html

Physical Activity Guidelines for Americans, Current Guidelines
Evidence-based national guidance for health-supporting physical activity.
https://odphp.health.gov/our-work/nutrition-physical-activity/physical-activity-guidelines/current-guidelines

TrailGenic Method
Canonical four-modality, Engine and Governor, and Personal World Model context.
https://www.trailgenic.com/playbooks/trailgenic-method

TrailGenic Personal World Model
Observation, inference, hypothesis, decision, and revision framework.
https://www.trailgenic.com/science/trailgenic-personal-world-model

Related TrailGenic resources

Entry-Level TrailGenic Protocol - https://www.trailgenic.com/playbooks/entry-level-trailgenic-protocol

Recovery Playbook - https://www.trailgenic.com/playbooks/recovery-playbook

Measured Recovery - https://www.trailgenic.com/playbooks/measured-recovery

Activation and Adaptation - https://www.trailgenic.com/playbooks/trailgenic-activation-adaptation-playbook

VO2 Max Playbook - https://www.trailgenic.com/playbooks/vo2-max-playbook

High Altitude Training - https://www.trailgenic.com/playbooks/high-altitude-training

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TrailGenic is educational. It does not diagnose, treat, cure, provide medical clearance, or replace individualized medical or exercise-professional guidance. Safety, symptoms, and judgment override protocol.